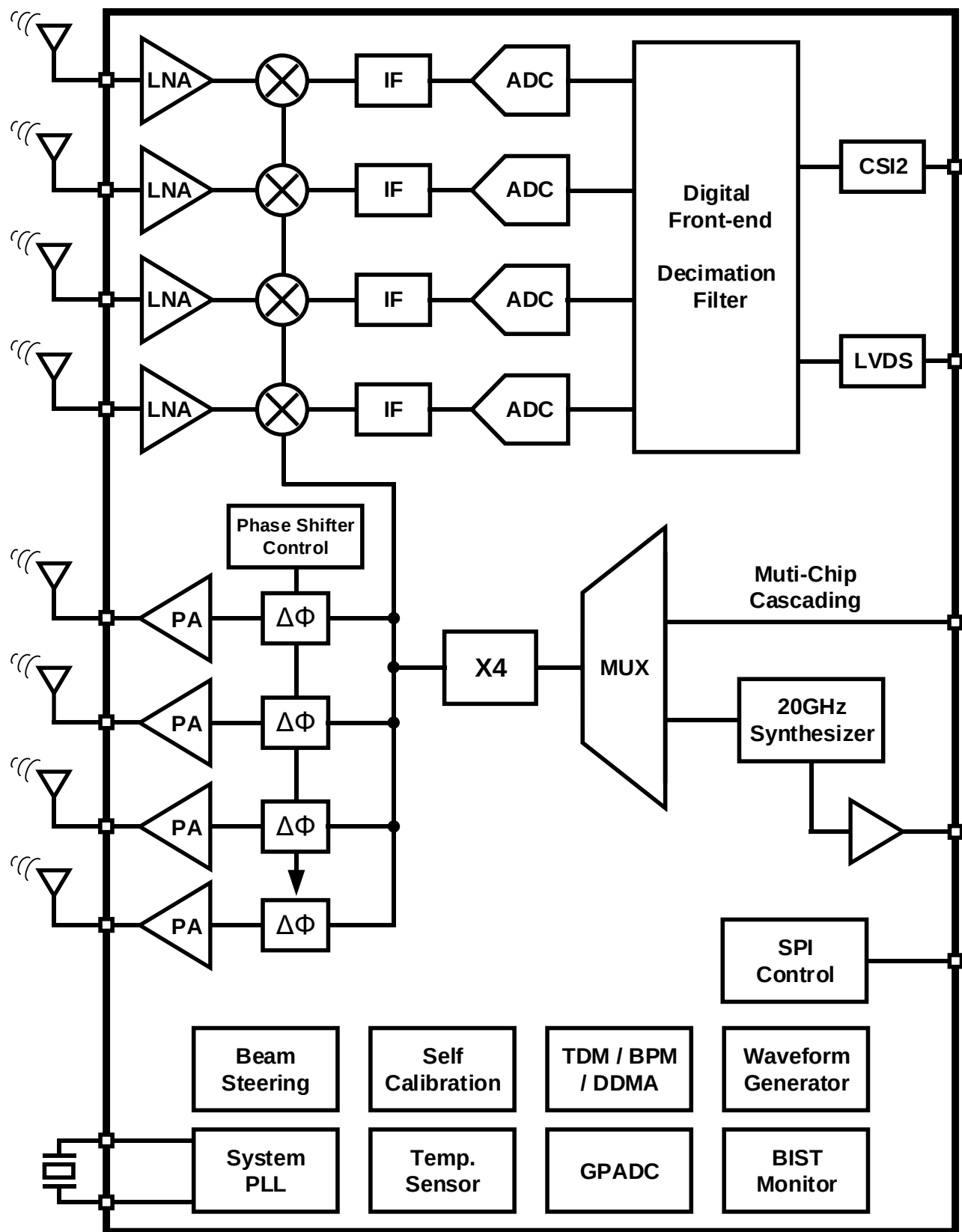


Cheetah4401M 4T4R 76-81GHz Radar Transceiver

1. Key Features

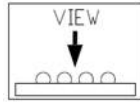
- FMCW transceiver
 - Integrated FMCW PLL, transmitter, receiver, baseband and ADC
 - 76- to 81-GHz coverage with 5 GHz available bandwidth
 - 4 receive channels
 - 4 transmit channels
 - Ultra-accurate chirp engine based on Fractional-N PLL
 - TX power: 14.5dBm (with package loss)
 - RX noise figure: 12.5 dB (without I/Q)
 - Phase noise at 1 MHz : – 99 dBc/Hz (76GHz to 77GHz);
 - Phase noise at 1 MHz : – 96 dBc/Hz (77GHz to 81GHz);
 - Arbitrary waveform generator for FMCW PLL
 - IF provide 10M/15M/20M/25M option
 - HPF provide 250kHz to 1MHz 3dB corner option (~50kHz per step)
 - Support 20GHz LO Cascading mode
 - Support TDM/BPM/DDMA MIMO radar function
 - Precise 6-bits phase shifter inside
- Built-in calibration and self-test and monitoring mechanism
 - Self-calibrating system across frequency and temperature
 - Calibrate and monitoring overall system during frame idle time
- Host interface
 - Control interface with external processor over SPI
 - Data interface with external processor over MIPI D-PHY and CSI2
 - LVDS high speed IO interface
 - Interrupts for Fault Reporting
- Power management
 - Built-in LDO Network for enhanced PSRR
 - I/Os support voltage from 1.8 V to 3.3 V
- Clock source
 - Supports externally driven clock (square/sine) at 40 MHz
 - Supports 40 MHz crystal connection with load capacitors
- Easy hardware design
 - 0.45-mm pitch, 265-pin 8.65 mm × 8.65 mm ewlb BGA package
 - Easy assembly and low cost PCB design
- Supports Grade1 automotive temperature operating range

2. Cheetah 4T4R Functional Block Diagram

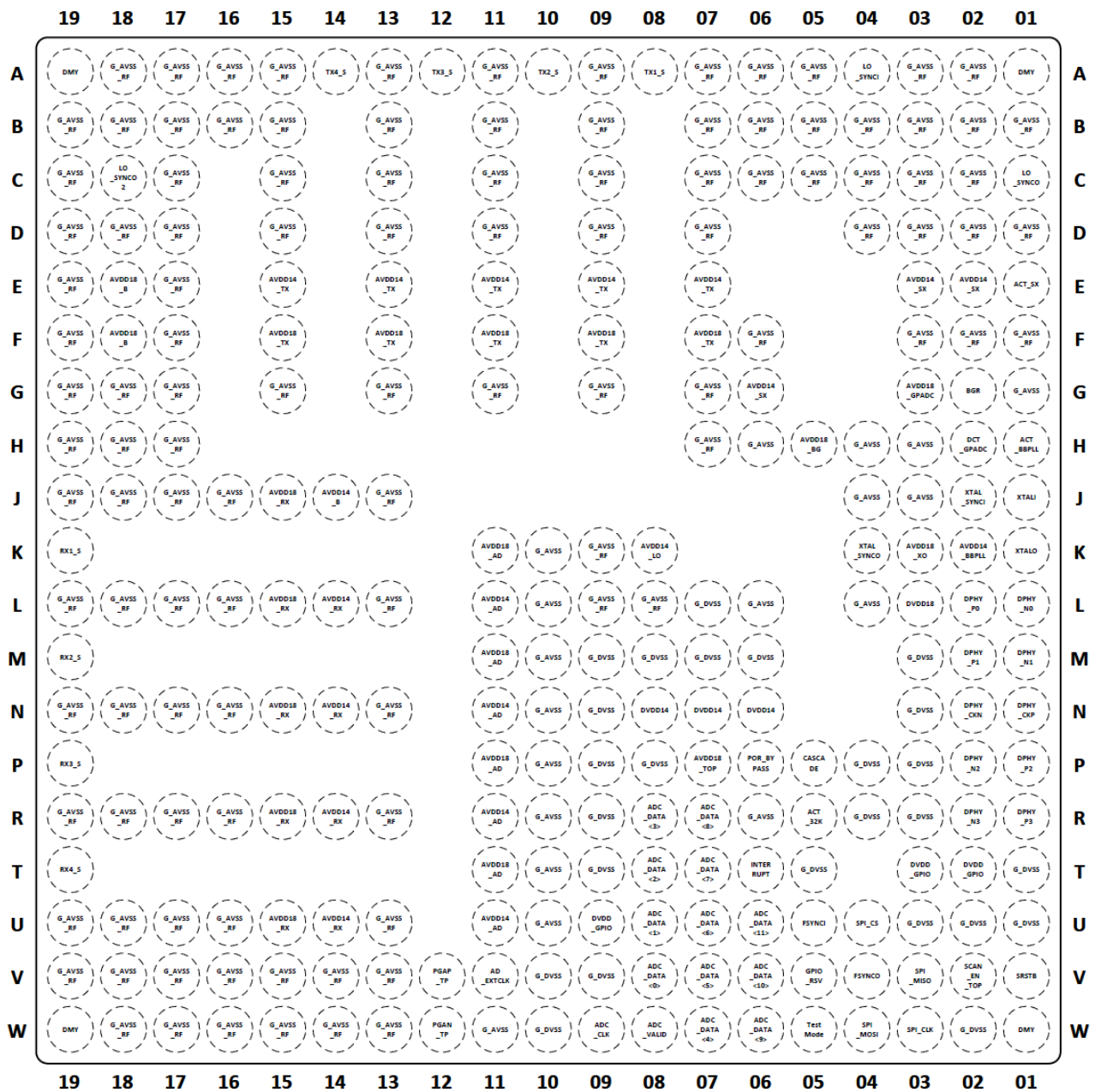


3. Cheetah 4T4R Pin Diagram

0.45-mm pitch, 265-pin 8.65 mm × 8.65 mm ewlb BGA package



Note : All pin diagram is face-up view



4. Cheetah 4T4R Pin Descriptions

RF Port :

Cheetah BGA mapping					
BGA Type	BGA_Nmae	# BGA	Max Current	Tunable-V Range	Comment
RF Port	TX1_S	1	N/A	N/A	80GHz RF TX Signal ; Connect to Antenna
RF Port	TX2_S	1	N/A	N/A	80GHz RF TX Signal ; Connect to Antenna
RF Port	TX3_S	1	N/A	N/A	80GHz RF TX Signal ; Connect to Antenna
RF Port	TX4_S	1	N/A	N/A	80GHz RF TX Signal ; Connect to Antenna
RF Port	RX1_S	1	N/A	N/A	80GHz RF RX Signal ; Connect to Antenna
RF Port	RX2_S	1	N/A	N/A	80GHz RF RX Signal ; Connect to Antenna
RF Port	RX3_S	1	N/A	N/A	80GHz RF RX Signal ; Connect to Antenna
RF Port	RX4_S	1	N/A	N/A	80GHz RF RX Signal ; Connect to Antenna
Sub-Total	RF Port BGA	8			

Power Pin :

Cheetah BGA mapping					
BGA Type	BGA_Nmae	# BGA	Max Current	Tunable-V Range	Comment
RF Power BGA	AVDD14_TX	5	200mA x 4	1.4V to 1.8V ; def. = 1.4V	RF-TX 1.4V clean power domain
RF Power BGA	AVDD18_TX	5	100mA x 4	1.8V x 4	RF-TX 1.8V clean power domain
RF Power BGA	AVDD14_RX AVDD14_B	5	60mA x5	1.4V to 1.8V ; def. = 1.4V	RF-RX 1.4V clean power domain
RF Power BGA	AVDD18_RX	5	15mA x4	1.8V	RF-RX 1.8V clean power domain
RF Power BGA	AVDD18_B	2	10mA x2	1.8V	RF-BIST 1.8V clean power domain
RF Power BGA	AVDD14_LO	1	100mA	1.4V to 1.8V ; def. = 1.4V	RF-LO 1.4V clean power domain
RF Power BGA	AVDD14_SX	3	80mA 50mA 20mA	1.4V to 1.8V ; def. = 1.4V	RF-SX 1.4V clean power domain
RF Power BGA	AVDD18_XO	1	20mA	1.8V	RF-XO 1.8V power domain
RF Power BGA	AVDD14_BBPLL	1	1mA 9mA 15mA	1.4V to 1.8V ; def. = 1.4V	RF-BBPLL 1.4V power domain
RF Power BGA	AVDD18_BG	1	9mA	1.8V	RF-BG 1.8V power domain
RF Power BGA	AVDD18_GADC	1	1mA	1.8V	RF-GPADC 1.8V power domain
RF Power BGA	AVDD18_TOP	1	1mA	1.8V	RF-TOP 1.8V power domain
RF Power BGA	AVDD18_AD	4	5mA x4	1.8V	RF-ADC 1.8V power domain
RF Power BGA	AVDD14_AD	4	40mA x4	1.4V to 1.8V ; def. = 1.4V	RF-ADC 1.4V power domain
Digital Power BGA	DVDD18	1	20mA	1.8V	Digital-DPHY 1.8V dirty power domain
Digital Power BGA	DVDD14	3	60mA Total	1.4V to 1.8V ; def. = 1.4V	Digital-DPHY/RFDIG 1.4V dirty power domain
Digital Power BGA	DVDD_GPI0	3	50mA	1.8V to 3.3V	Digital-GPI0 ; 1.8V~3.3V ; dirty power domain
Sub-Total	Power BGA	46			

GND Pin :

Cheetah BGA mapping					
BGA Type	BGA_Nmae	# BGA	Max Current	Tunable-V Range	Comment
RF GND BGA	G_AVSS_RF	112	< 2000mA	0V	Separated GND ; RF subblock = weak-connection @ Chip ; TX = Separated @ PKG ; = All connect @ PCB
RF GND BGA	G_AVSS	18	<250mA	0V	Separated GND
Digital GND BGA	G_DVSS	25	<200mA	0V	Separated GND
Sub-Total	GND BGA	155			

IO Pin :

Cheetah BGA mapping					
BGA Type	BGA_Nmae	# BGA	Max Current	Tunable-V Range	Comment
Digital IO	SPI_CLK	1	N/A	1.8V to 3.3V	DVDD_GPIO Domain(1.8~3.3V) ; max=40MHz ; GPIO other function @ Digital IO_GPIO part
Digital IO	SPI_CS	1	N/A	1.8V to 3.3V	DVDD_GPIO Domain(1.8~3.3V) ; SPI select ; GPIO other function @ Digital IO_GPIO part
Digital IO	SPI_MOSI	1	N/A	1.8V to 3.3V	DVDD_GPIO Domain(1.8~3.3V) ; SPI input ; GPIO other function @ Digital IO_GPIO part
Digital IO	SPI_MISO	1	N/A	1.8V to 3.3V	DVDD_GPIO Domain(1.8~3.3V) ; SPI output ; GPIO other function @ Digital IO_GPIO part
Digital IO	ADC_CLK	1	N/A	1.8V to 3.3V	DVDD_GPIO Domain(1.8~3.3V) ; for RF ADC test mode ; up to 120M
Digital IO	ADC_VALID	1	N/A	1.8V to 3.3V	DVDD_GPIO Domain(1.8~3.3V) ; for RF ADC test mode ; up to 120M
Digital IO	ADC_DATA<11:0>	12	N/A	1.8V to 3.3V	DVDD_GPIO Domain(1.8~3.3V) ; for RF ADC test mode ; up to 120M ; GPIO other function @ Digital IO_GPIO part
Digital IO	INTERRUPT	1	N/A	1.8V to 3.3V	DVDD_GPIO Domain(1.8~3.3V) ; 40MHz Clock Domain ; GPIO other function @ Digital IO_GPIO part
Digital IO	FSYNCO	1	N/A	1.8V to 3.3V	DVDD_GPIO Domain(1.8~3.3V) ; 40MHz Clock Domain ; GPIO other function @ Digital IO_GPIO part
Digital IO	FSYNCI	1	N/A	1.8V to 3.3V	DVDD_GPIO Domain(1.8~3.3V) ; 40MHz Clock Domain ; GPIO other function @ Digital IO_GPIO part
Digital IO	SRSTB	1	N/A	1.8V to 3.3V	DVDD_GPIO Domain(1.8~3.3V) ; Reset TOPDIG/RFDIG Power-On-Seq (Reset RG is an optional function)
Digital IO	TestMode	1	N/A	1.8V to 3.3V	DVDD_GPIO Domain(1.8~3.3V) ; testing mode enable
Digital IO	SCAN_EN_TOP	1	N/A	1.8V to 3.3V	DVDD_GPIO Domain(1.8~3.3V) ; scan mode enable
Digital IO	GPIO_RSV	1	N/A	1.8V to 3.3V	DVDD_GPIO Domain(1.8~3.3V) ; GPIO Reserve
AIO-AD	AD_EXTCLK	1	N/A	1.8V	ADC Testing input clock, up to 120MHz
AIO-DPHY	DPHY_CK_P	1	N/A	N/A	DPHY output clock
AIO-DPHY	DPHY_CK_N	1	N/A	N/A	DPHY output clock
AIO-DPHY	DPHY_P0	1	N/A	N/A	DPHY output data
AIO-DPHY	DPHY_P0	1	N/A	N/A	DPHY output data
AIO-DPHY	DPHY_P1	1	N/A	N/A	DPHY output data
AIO-DPHY	DPHY_P1	1	N/A	N/A	DPHY output data
AIO-DPHY	DPHY_P2	1	N/A	N/A	DPHY output data
AIO-DPHY	DPHY_P2	1	N/A	N/A	DPHY output data
AIO-DPHY	DPHY_P3	1	N/A	N/A	DPHY output data
AIO-DPHY	DPHY_P3	1	N/A	N/A	DPHY output data
AIO-LO	LO_SYNCI	1	N/A	N/A	LO sync in/out in cascade mode ; 50 ohm ; high speed up to 20.25GHz
AIO-LO	LO_SYNCO	1	N/A	N/A	LO sync in/out in cascade mode ; 50 ohm ; high speed up to 20.25GHz
AIO-LO	LO_SYNCO2	1	N/A	N/A	LO sync in/out in cascade mode ; 50 ohm ; high speed up to 20.25GHz
AIO-ACT	ACT_SX	1	25mA	N/A	50 ohm ; high speed up to 3GHz ; Testing pin
AIO-ACT	ACT_BBPLL	1	25mA	N/A	50 ohm ; high speed up to 6GHz ; Testing pin
AIO-ACT	ACT_32K	1	N/A	N/A	1.1V 32K/64K output for testing
AIO-ACT	PGAN_TP	1	N/A	N/A	1.1V domain ; Need transformer ; for AFE output testing point or ADC input testing point
AIO-ACT	PGAP_TP	1	N/A	N/A	1.1V domain ; Need transformer ; for AFE output testing point or ADC input testing point
AIO	BGR	1	100uA	N/A	Connect 24K <1% resistor ; BG calibration usage ; 1.8V domain
AIO	POR_BYPASS	1	N/A	0 or 1.8V	Bypass POR option ; 1.8V=bypass ; 0V=normal mode ; 1.6M weak tie-low resistor inside
AIO	CASCADE	1	N/A	0 or 1.8V	Cascade mode strap pin ; 1.8V=cascade mode ; 0V=normal mode ; 1.6M weak tie-low resistor inside
AIO	XTALI	1	N/A	N/A	XO gate ; connect to 40M / 8pF / ESR<30 Crystal BOM and XO-buffer in chip ; need external cap = 8pF
AIO	XTALO	1	N/A	N/A	XO drain ; connect to 40M / 8pF / ESR<30 Crystal BOM ; need external cap = 16pF
AIO	XTAL_SYNCI	1	N/A	N/A	40M clock sync in/out in cascade mode
AIO	XTAL_SYNCO	1	N/A	N/A	40M clock sync in/out in cascade mode
AIO-DCT	DCT_GPADC	1	N/A	N/A	1.8V domain ; Probing ; for DC test
Sub-Total	IO	52			
Dmy Ball	Dmy Ball	4			
Total	Power/GND/IO	265			

5. RF Specification

PARAMETER		MIN	TYP	MAX	UNIT
Receiver	Noise Figure at 1MHz (w/o blocker tone, SSB) ⁽¹⁾		12.5		dB
	1-dB Compression Point (out of band)(10kHz) ⁽²⁾		−8		dBm
	Maximum Gain (in band)		54		dB
	Gain Range		20		dB
	Gain Step Size		2		dB
	IF Bandwidth ⁽³⁾			25	MHz
	ADC Sampling Rate (decimation with factor=2x)			53.3	Msp/s
	ADC Resolution		12		Bits
	Return Loss (S11)		<−10		dB
	Gain Flatness (76GHz to 81GHz)		2		dB
	Gain Mismatch Variation (over temperature)		±0.5		dB
	Phase Mismatch Variation (over temperature)		±3		°
	IIP2 (Out-Band) (10kHz & 8MHz)		27		dBm
	IIP2 (In-Band) (8MHz & 9MHz)		16		dBm
	Idle RX Channel Spurs		−90		dBFS
	RX to RX Channel Isolation		30		dB
Transmitter	Output Power		14.5		dBm
	Output Noise Level at 1MHz Offset		−145		dBc/Hz
	Output Power Level Flatness (76GHz to 81GHz)		3		dB
	TX to RX Channel Isolation		30		dB
	TX Channel Spurs		−80		dBc
LO	Frequency Range	76		81	GHz
	Ramp Rate			266	MHz/us
	Phase Noise at 1MHz Offset (76GHz to 77GHz)		−99		dBc/Hz
	Phase Noise at 1MHz Offset (77GHz to 81GHz)		−96		dBc/Hz
	Spur Level		<−45		dBc/Hz
SYNC IO (LO / 4)	SYNC-IN Return Loss		−9		dB
	SYNC-IN Input Power Level	−6			dBm
	SYNC-OUT Return Loss		−9		dB
	SYNC-OUT Output Power Level	3	7		dBm

Note:

- (1) SSB Noise figure is DSB case add 3dB, in our case DSB Noise figure is 10dB maximum, SSB Noise figure is 13dB maximum w/o IQ.
- (2) 1-dB Compression Point (Out Of Band) is measured by feeding a 10kHz continuous wave tone.
- (3) The analog IF include high-pass/low-pass filtering, low-pass corner can set by 10/15/20/25 MHz.

6. Power Summary

6.1. Power consumption

Maximum Current Rating at Power Terminals ⁽¹⁾

Parameter	SUPPLY NAME	Description	MAX Current	UNIT
AVDD14	AVDD14_TX AVDD14_RX AVDD14_B AVDD14_SX AVDD14_LO	Internal LDO driven by 1.4V to 1.8V	1350	mA
AVDD18	AVDD18_TX AVDD18_RX AVDD18_B AVDD18_XO AVDD18_BG AVDD18_GPADC AVDD18_TOP	Internal LDO driven by 1.8V	500	
DVDD14	DVDD14 AVDD14_BBPLL AVDD14_AD	Internal LDO driven by 1.4V to 1.8V	250	
DVDD18	DVDD18 AVDD18_AD	Internal LDO driven by 1.8V	10	
DVDD_GPIO	DVDD_GPIO	Internal FPIO driven by 1.8V to 3.3V	50	

Note:

(1) The specified current values are at typical supply voltage level.

Average Power Consumption at Power Terminals

Parameter	Condition	Description	Typical Power	UNIT
Average Power Consumption	4T / 4R	100% Duty Cycle	2.3	W
	4T / 4R	50% Duty Cycle	1.3	
	4T / 4R	25% Duty Cycle	0.8	
	3T / 4R	50% Duty Cycle	1.0	
	3T / 4R	25% Duty Cycle	0.65	

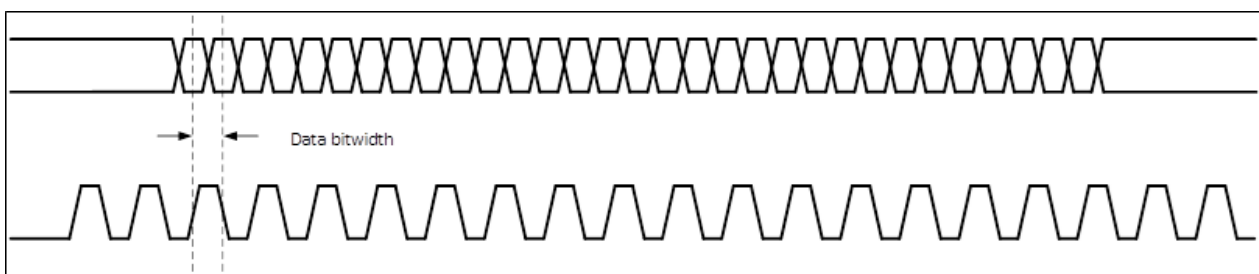
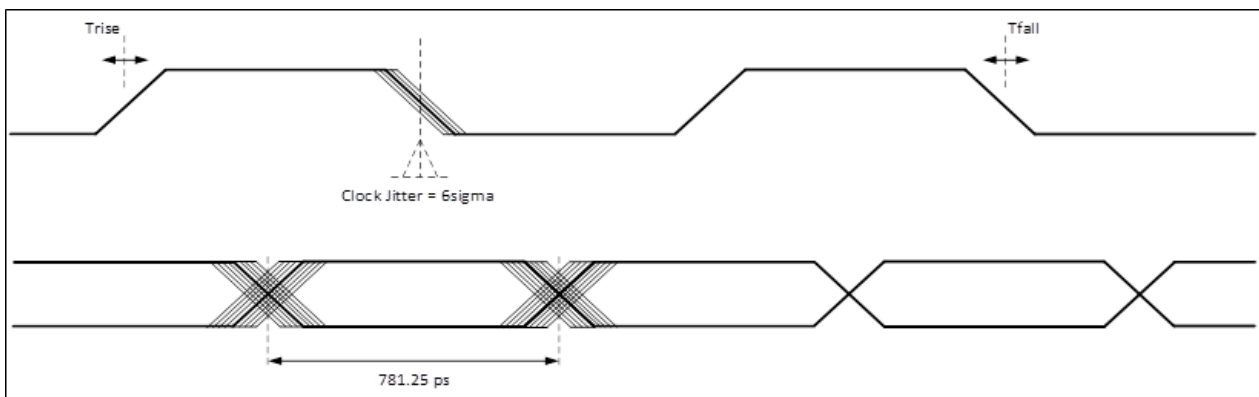
6.1. Power on sequence

Power on sequence is VDDIO, 1.4V, 1.8V. The time interval should be greater than 1 millisecond.

7. IO Timing

7.1. LVDS

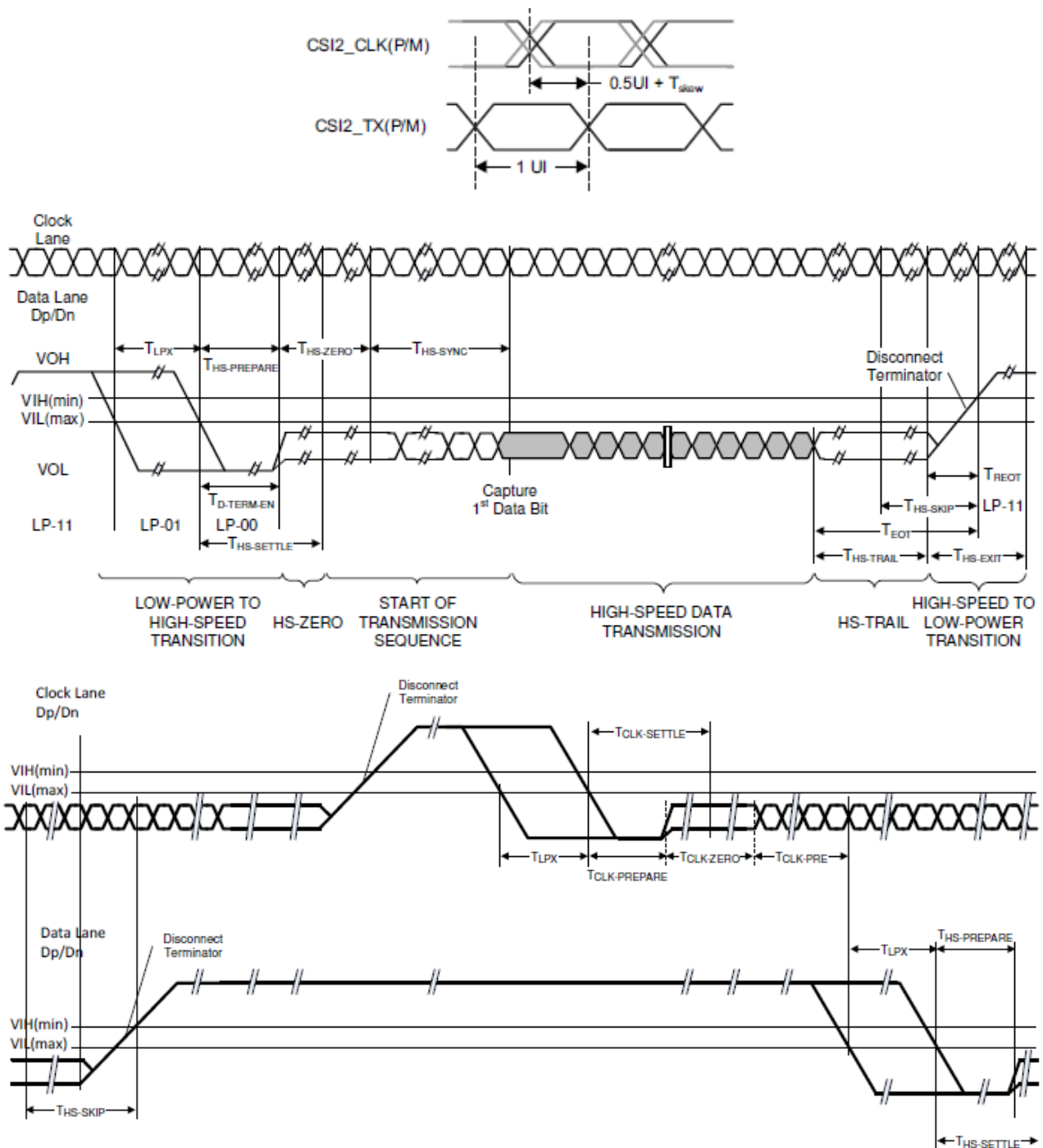
PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
Duty Cycle Requirements	max 1pF lumped capacitive load on LVDS lanes	48%		52%	
Output Differential Voltage	peak-to-peak single ended with 100Ω resistive load between differential pairs	250		450	mV
Output Offset Voltage		925		1025	mV
Trise and Tfall	20% to 80%, 1280Mbps		330		ps
Jitter(pk-pk)	1280Mbps		80		ps



7.2. CSI2

PARAMETER	Description	MIN	TYP	MAX	UNIT
HSTX					
$HSTX_{DBR}$	Data bit rate	160		1280	Mbps
f_{clk}	DDR clock frequency	80		640	MHz
$\Delta V_{CMTX(LF)}$	Common-level variations between 80-450MHz			25	mV _{PEAK}
$\Delta V_{CMTX(HF)}$	Common-level variations above 450MHz			15	mV _{RMS}
Trise and Tfall	20% to 80% rise time and fall time below 1Gbps			0.3	UI
	20% to 80% rise time and fall time above 1Gbps			0.35	UI
	20% to 80% rise time and fall time below 1Gbps	150			ps
	20% to 80% rise time and fall time above 1Gbps	100			ps
V_{CMTX}	HS transmit static sommon-level voltage	150	200	250	mV
$ \Delta V_{CMTX(1,0)} $	V_{CMTX} mismatch when output is Differential-1 or Differential-0			5	mV
$ V_{OD} $	HS transmit differential voltage	140	200	270	mV
$ \Delta V_{OD} $	V_{OD} mismatch when output is Differential-1 or Differential-0			14	mV
Z_{OS}	Single ended output impedance	40	50	62.5	Ω
ΔZ_{OS}	Single ended output impedance mismatch			10	%
LPTX					
V_{OH}	Thevenin output high level	1	1.1	1.2	V
V_{OL}	Thevenin output low level	-50		50	mV
Z_{OLP}	Output impedance of LP transmitter	110			Ω
DATA-CLOCK Timing Specification					
UINOM	Nominal Unit Interval	0.78125		6.25	ns
UIINST,MIN	Minimum instantaneous Unit Interval	0.53			ns
TSKEW[TX]	Data to clock skew measured at transmitter	-0.15		0.15	UIINST,MIN

CSI2 Timing SPECIFICATION			
$T_{CLK-PRE}$	Time that the HS clock shall be driven by the transmitter before any associated data lane beginning the transition from LP to HS mode	8	ns
$T_{CLK-PREPARE}$	Time tha transmitter drives the clock lane LP-00line state immediately before the HS-0 line state starting the HS transmission	38 95	ns
$T_{CLK-PREPARE} + T_{CLK-ZERO}$	$T_{CLK-PREPARE}$ + time that the transmitter drives the HS-0 state before starting the clock	300	ns
T_{EOT}	Transmitted time interval from the start of $T_{HS-TRAIL}$ or $T_{CLK-TRAIL}$ to the start of the LP-11 state following a HS burst	105ns+ 12*UI	ns
$T_{HS-PREPARE}$	Time that the transmitter drives the data lane LP-00line state immediately before the HS-0 line state starting the HS transmission	40ns+4*UI 85ns +6*UI	ns
$T_{HS-PREPARE} + T_{HS-ZERO}$	$T_{HS-PREPARE}$ + time that the transmitter drives the HS-0 state prior to transmitting the Sync sequence	145ns+ 10*UI	ns
$T_{HS-EXIT}$	Time that the transmitter drives LP-11 following a HS burst	100	ns
$T_{HS-TRAIL}$	Time that the transmitter drives the flipped differential state after last payload data bit of a HS transmission burst	max(8*UI 60ns+ 4*UI)	ns
T_{LPX}	Transmitted length of any low-power state period	50	ns



(1) The HS to LP transition of the CLK does not actually take place since the CLK is always ON in HS mode.

7.3. SPI Bus

As a slave device, Cheetah communicates with master by standard 4 lines SPI bus. Following table shows the meaning of the signals.

Table SPI signals description

signal	Function description
SPI_CLK	SPI clock : $\leq 10\text{MHz}$
SPI_CS	SPI chip selection signal : active low
MISO	Master in slave out (cheetah out)
MOSI	Master out slave in (cheetah in)

SPI timing is illustrated below.

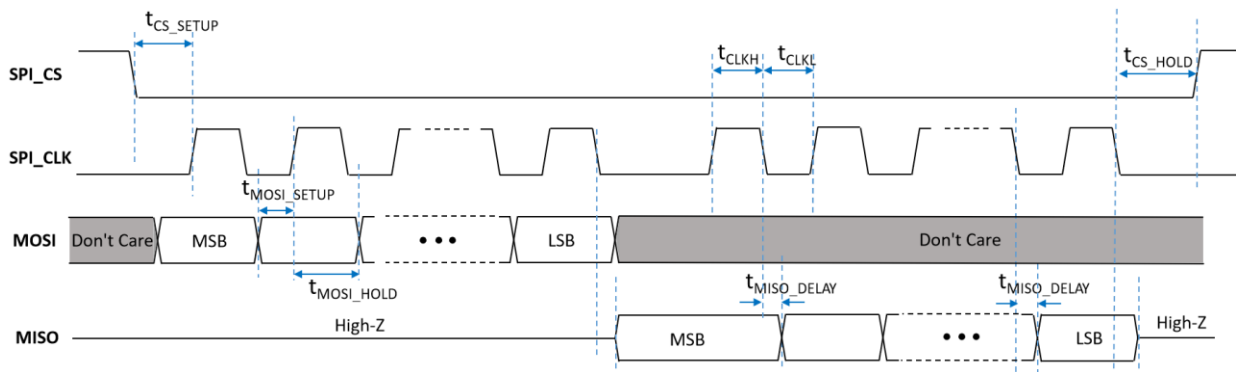
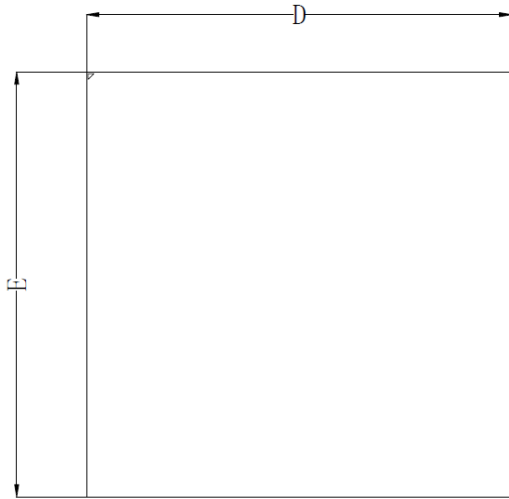


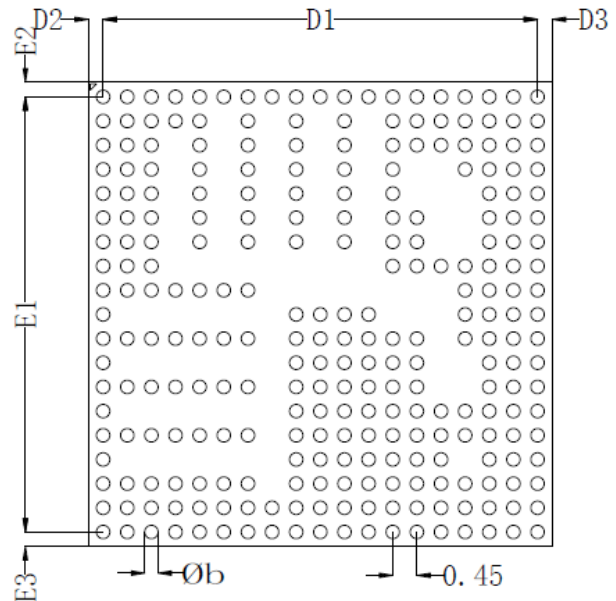
Table SPI Timing Constraints

Symbol	Description	Condition	Min	Max	Unit
f_{clk}	Clock frequency	Max Load : 25pF		10	MHz
t_{CLKL}	Clock low time		20		ns
t_{CLKH}	Clock high time		20		ns
t_{MOSI_SETUP}	MOSI setup time		20		ns
t_{MOSI_HOLD}	MOSI hold time		20		ns
t_{MISO_DELAY}	MISO output delay	Load=25pF		30	ns
		Load=250pF		40	ns
t_{CS_SETUP}	CS setup time		20		ns
t_{CS_HOLD}	CS hold time		40		ns

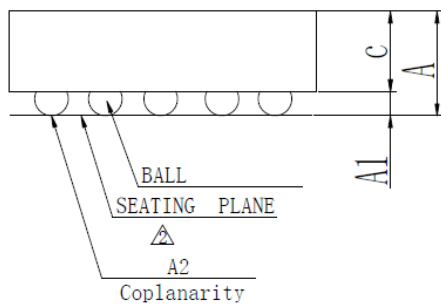
8. Packaging



Top View (Bump Face Down)



Bottom View (Bump Face up)



Symbol	Dimension in mm		
	MIN	NOM	MAX
A	0.6600	0.7000	0.7400
A1	0.1650	0.1950	0.2250
A2		0.1000	
C	0.4800	0.5050	0.5300
D	8.6000	8.6500	8.7000
E	8.6000	8.6500	8.7000
b	0.2350	0.2650	0.2950
D1		8.1000	
E1		8.1000	
D2		0.2750	
E2		0.2750	
D3		0.2750	
E3		0.2750	
Bump number	265		

1. CONTROLLING DIMENSION : MICROMETER.
2. PRIMARY DATUM C AND SEATING PLANE ARE DEFINED BY THE SPHERICAL CROWNS OF THE COPPER PILLAR BUMPS.
3. DIMENSION b IS MEASURED AT THE MAXIMUM COPPER PILLAR BUMP DIAMETER, PARALLEL TO PRIMARY DATUM C.
4. MINIMUM BUMP PITCH 0.450mm.
5. VIEW OF DIE IS BASED ON WAFER NOTCH AT THE BOTTOM.
6. KERF SIZE:0.070mm